

Rana Dolui

Full Stack Engineer | Go | Python | TypeScript | Distributed Systems | Cloud | AI

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SUMMARY

Full Stack Engineer with expertise in building scalable web applications, high-performance backend systems, and AI-powered platforms. Experienced in developing responsive frontend applications with React, Next.js, and React Native, alongside designing distributed microservices, REST/GraphQL APIs, and cloud-native backend infrastructure using Go, Python, FastAPI, PostgreSQL, Redis, Docker, Kubernetes, and AWS. Passionate about performance optimization, system design, and open-source software.

SKILLS

- **Languages:** JavaScript, TypeScript, Python, Go, C/C++
- **Frameworks & Libraries:** Next.js, React, React Native, Svelte, Express.js, FastAPI, Django, Fiber, GraphQL, gRPC
- **Databases & Baas:** PostgreSQL, MySQL, MongoDB, SQLite, Firebase, Redis, Supabase, ScyllaDB, Cassandra
- **DevOps & Cloud:** Docker, Nginx, Kubernetes, AWS, GCP, CI/CD, Git, Kafka, Cloudflare,

WORK EXPERIENCE

React Native Developer HumanEra Intelligence Intern	SEP 2024 - DEC 2024
Engineered a cross-platform social media application using React Native and Expo, implementing an offline-first architecture with a local database for improved performance and user experience. Designed background processing workflows and configured platform-specific permissions for Android and iOS. Contributed to backend development by building and maintaining GraphQL APIs with Express.js and Prisma, supporting scalable and reliable application services.	
Full stack Developer Saciva Full Time	JAN 2025 - NOV 2025
Built and maintained production-ready full-stack applications using modern frontend frameworks and backend technologies. Developed responsive user interfaces, designed scalable REST APIs, optimized PostgreSQL databases, and implemented real-time notifications. Built a recommendation system powered by Typesense and a custom classification model. Designed microservices, automated CI/CD pipelines with GitHub Actions, and deployed applications on AWS (EC2, S3, and RDS).	
Python Developer Telo Part time	DEC 2025 - Feb 2026
Designed and developed production-grade backend services and AI agent systems using FastAPI, PostgreSQL, and Redis. Integrated LangChain and OpenAI to build intelligent automation workflows, document processing, and AI-powered features. Implemented secure APIs, optimized application performance, and streamlined deployments through Docker and GitHub Actions on VPS infrastructure.	
Freelancing Part time	NOV 2025 - PRESENT
I work as a freelance backend developer, building and fixing backend systems using FastAPI, Fiber, Express.js, and Hono.js. I handle large projects, performance issues, and secure API design, and I'm comfortable working with messy codebases. I also work with PostgreSQL, Redis, Docker, and Kubernetes, managing everything from development to deployment on VPS or cloud.	

OPEN SOURCE PROJECTS

- FlashORM: High-Performance Multi-Database ORM (Go)**
- A high-performance ORM in Go with sqlc-like functionality, focused on speed and type safety.
 - Designed type-safe code generation supporting Go, JS/TS, Python, java.
 - Implemented support for multiple databases: MySQL, PostgreSQL, SQLite, Cassandra, scylladb.
 - Developed a plugin-based architecture and built-in Studio for schema management, enabling database operations with or without using the ORM.
- AstraAPI: High-Performance Backend Framework for Python (C++)**
- Built a high-performance backend framework for Python, implemented in C++ for maximum speed and efficiency.
 - Developed a custom native HTTP server, eliminating ASGI/Starlette dependencies with a fully C++-based runtime.
 - Designed a lightweight multi-worker architecture, **using ~100 MB RAM for 3 workers (vs. ~300–500 MB in typical Python setups).**
 - Achieved **84–88% worker efficiency**, optimizing concurrency and resource utilization.
 - Engineered the system to **handle ~200K requests/second per worker**, enabling high scalability and throughput.

PROJECTS

- **AssumioAI**: Developed a chat-first AI platform featuring a multi-agent pipeline that analyzes market size, competition, feasibility, risks, and execution strategy to validate startup ideas. Built with FastAPI, Next.js, PostgreSQL, Pinecone, and LangChain, processing 50+ data sources and generating comprehensive startup validation reports in under 60 seconds.
- **OmniAI**: Architected and developed a distributed AI platform that enables users to chat with documents and websites using Retrieval-Augmented Generation (RAG). Designed distributed ingestion pipelines for document uploads, OCR, URL crawling, vector embeddings, and semantic search. Built scalable microservices communicating over gRPC and RabbitMQ, using Next.js, Go/Fiber, FastAPI, PostgreSQL, Redis, Pinecone, Docker, LangChain, and RabbitMQ.
- **AiResume**: AI-powered resume SaaS for generating resumes and cover letters, analyzing CVs, and producing ATS-friendly results. Includes an admin panel, user and plan management, Razorpay payments, and full role-based access control. Built with Next.js, Firebase, OpenAI, FastAPI, Razorpay, MySQL, Docker, and Nginx.
- **Proxyon**: Developed a website that allows users to configure an API gateway or load balancer by connecting nodes (like n8n), enabling setup with just a few clicks. Built with Next.js, React Flow, TailwindCSS, Framer Motion, Go, Fiber, NATS, MongoDB, PostgreSQL, Docker, Nginx, S3, Terraform, and CoreDNS.
- **HireLens**: is an AI-powered platform that helps users improve their interview skills. Built with Next.js, Prisma, TypeScript, and Gemini, it offers personalized mock interviews, real-time feedback, and expert insights, making interview preparation more effective and confidence-boosting.
- **khelratna**: Developed a WebView-based cricket tournament platform for hosting local matches and tournaments, **serving 1,000+ registered users**. Built a comprehensive admin panel with role-based access control (RBAC) to manage users, tournament owners, captains, and organizers. Designed a **highly optimized backend capable of simultaneously live-streaming two matches to 600+ concurrent users on a 1 vCPU, 2 GB RAM VPS**. Built with Next.js, Socket.IO, Go, FastAPI, Supabase, Flutter, Redis, Docker, and Firebase.
- **Passman**: An ultra-secure password manager with end-to-end zero-knowledge encryption. It supports authenticator app-based verification, live session management, and a zero-trust mode that allows only a single active, IP-bound session to prevent session hijacking. Built with Svelte, FastAPI, Redis and PostgreSQL.